



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

In this project, a rival company to SpaceX uses SpaceX Falcon 9 rocket data to determine the rocket first stage landing successes and the cost of a launch.

Summary of methodologies

- Data Collection
 - Data wrangling
 - Exploratory data analysis with data visualization and SQL
 - Building an interactive map with Folium
 - Building Dashboard with Plotly Dash
 - Predictive analysis (classification)
- **Summary of all results**
 - Exploratory data analysis results
 - Interactive analytics demo in screenshots

Introduction

This capstone project is part of the IBM Data Science Professional Certificate. The goal of the project is to demonstrate proficiency in data science and machine learning techniques using a real-world data and to summarize the results in a report.

In this project, a rival company to SpaceX (i.e., SpaceY) uses SpaceX Falcon 9 rocket data to

- **determine the rocket first stage landing successes**
- **what factors can be used to predict this.**

The major parts of this report include data collection methodology, data wrangling, exploratory data analysis (EDA), interactive data visualization, machine learning (ML) classification model development, and model evaluation.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - REST API
 - Wikipedia
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Multiple models were tested and compared to determine best fit.

The wrangling of the collected data included cleaning, preparation for visualization and information extraction for usage in ML predictive models such as logistic regression, support vector machine (SVM), decision tree, and K-nearest neighbors (KNN).

In addition, exploratory data analysis (EDA) was performed using visualization and SQL. Lastly, Folium and Plotly Dash Python libraries were used in data representation and in the interactive visual analytics of the data.

Finally, predictive analysis was performed using classification models for predicting if the first stage of Falcon 9 rocket will land successfully using

Data Collection

- Data used in this project were collected from SpaceX Rest API and from Wikipedia launch table.

Data collection and visualization major steps:

Step 1:
Collect Data
from SpaceX API and
Convert data to
.JSON

Step 2:
Filter data to include Falcon 9 data,
assign data to dataframe and
dictionary, and export data to a csv file

Step 3:
Plot and visualize the
data

Data Collection – SpaceX API

Request response from SpaceX API using get request and convert data to .JSON file

Clean the data

Assign data to dictionary and data frame

Filter data to include only Falcon 9 launches and export data to a CSV

```
] 1 static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'

] 1 # convert the json result into a dataframe using json_normalize method
  2 data=pd.json_normalize(response.json())

1 # Lets take a subset of our dataframe keeping only the features we want and the flight number, and date_utc.
2 data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_utc']]
3 #
4 # We will remove rows with multiple cores because those are falcon rockets with 2 extra rocket boosters and rows that have multiple payloads in a single rocket.
5 data = data[data['cores'].map(len)==1]
6 data = data[data['payloads'].map(len)==1]
7
8 # Since payloads and cores are lists of size 1 we will also extract the single value in the list and replace the feature.
9 data['cores'] = data['cores'].map(lambda x : x[0])
10 data['payloads'] = data['payloads'].map(lambda x : x[0])
11
12 # We also want to convert the date_utc to a datetime datatype and then extracting the date leaving the time
13 data['date'] = pd.to_datetime(data['date_utc']).dt.date
14
15 # Using the date we will restrict the dates of the launches
16 data = data[data['date'] <= datetime.date(2020, 11, 13)]
```

```
[ ] 1 data_falcon9.head()
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
4	1	2010-06-04	Falcon 9	6123.547647	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0003	-80.577366	28.561857
5	2	2012-05-22	Falcon 9	525.000000	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0005	-80.577366	28.561857
6	3	2013-03-01	Falcon 9	677.000000	ISS	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0007	-80.577366	28.561857
7	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None	1.0	0	B1003	-120.610829	34.632093
8	5	2013-12-03	Falcon 9	3170.000000	GTO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B1004	-80.577366	28.561857

Data Collection - Scraping

Step 1: Perform HTTP get to request Falcon 9 HTML page and create BeautifulSoup object from HTML

Step 2: Extract all column/variable names from the HTML table header

Step3: Create a data frame by parsing the launch HTML tables

Step 4: export data into CSV file

```
[5] 1 # use requests.get() method with the provided static_url
    2 # assign the response to a object
    3 response = requests.get(static_url)
```

Create a BeautifulSoup object from the HTML response

```
[6] 1 # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
    2 soup = BeautifulSoup(response.content, 'html.parser')
```

Print the page title to verify if the BeautifulSoup object was created properly

```
[7] 1 # Use soup.title attribute
    2 soup.title
```

<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>

<

Data Wrangling

Step 1:

Load data from dataset_part1.csv file and calculate the number of launches on each site

Step 2:

Calculate the number and the occurrence of each orbit

Step3:

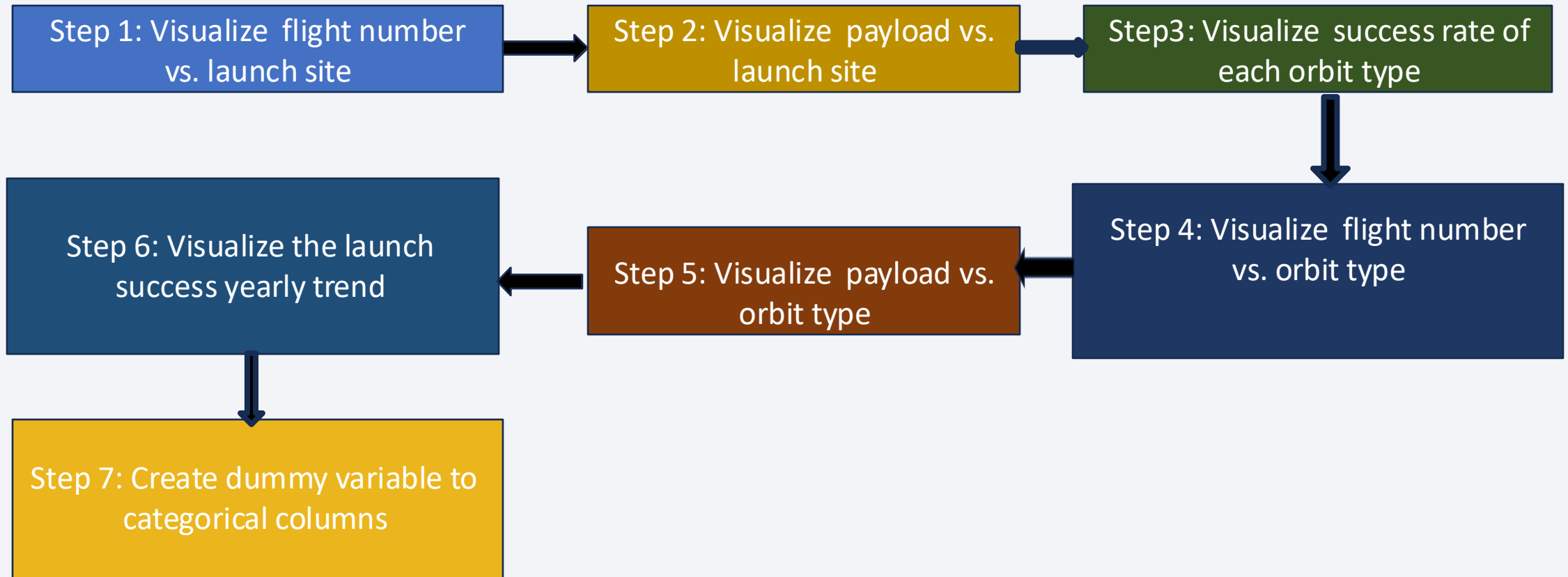
Calculate the number and occurrence of mission outcome of the orbits

Step 4:

Create a landing outcome label from outcome column and export data to a csv file

EDA with Data Visualization

Use Matplotlib and Seaborn for data visualization



EDA with SQL

- Step 1: Display the names of the unique launch sites in the space mission
- Step 2: Display 5 records where launch sites begin with the string 'CCA'
- Step 3: Display average payload mass carried by booster launched by NASA (CRS)
- Step 4: Display average payload mass carried by booster version F9 v1.1
- Step 5: List the date when the first successful landing outcome in ground pad was achieved
- Step 6: List the names of the boosters which have success in drone ship and mass > 4000 & <6000
- Step 7: List the total number of successful and failure mission outcomes
- Step 8: List the names of the booster_ve which have carried the max payload mass
- Step 9: List the records which display the month, failure landing, booster version ..etc.
- Step 10: Rank the count of landing outcomes or success

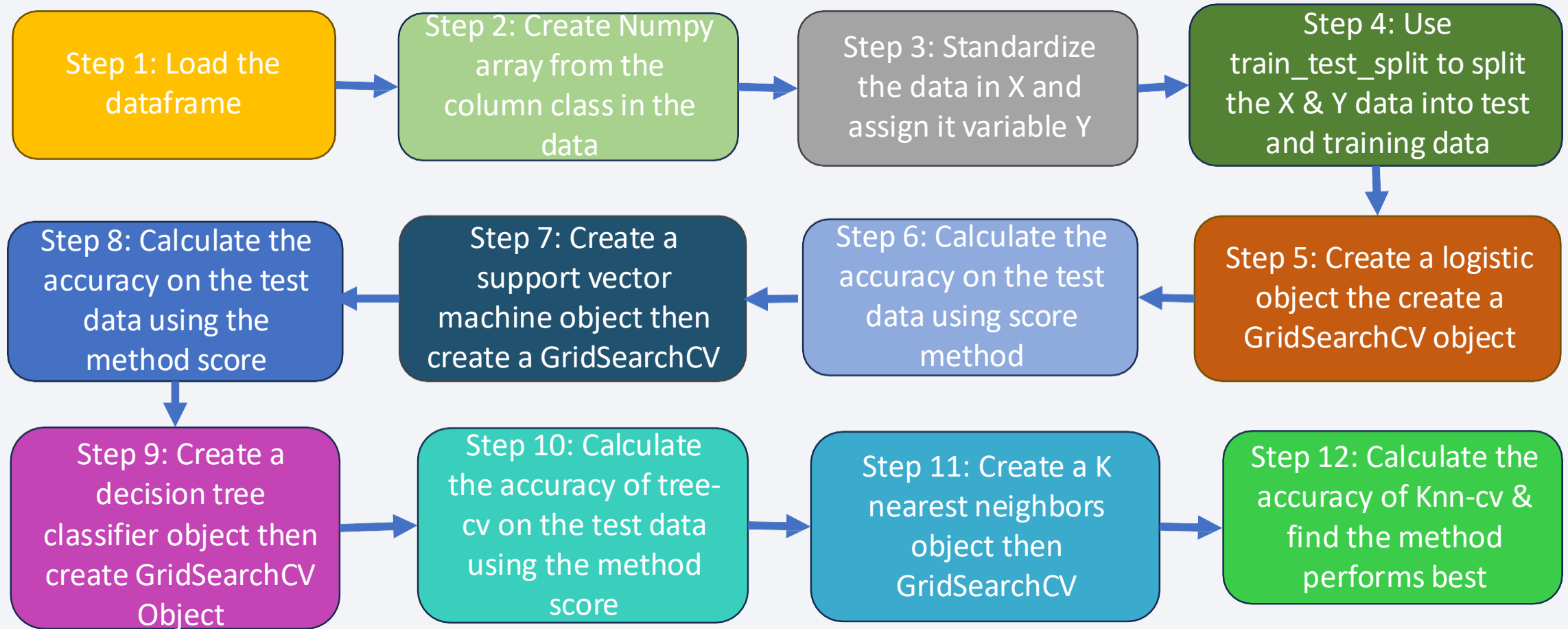
Build an Interactive Map with Folium

- Mark all launch sites on a map created using Folium by adding markers with circle , popup label and text label to each site using its longitude and latitude to show the geographical location
- Mark all the success/failed launches for each site on the map using colored markers

Build a Dashboard with Plotly Dash

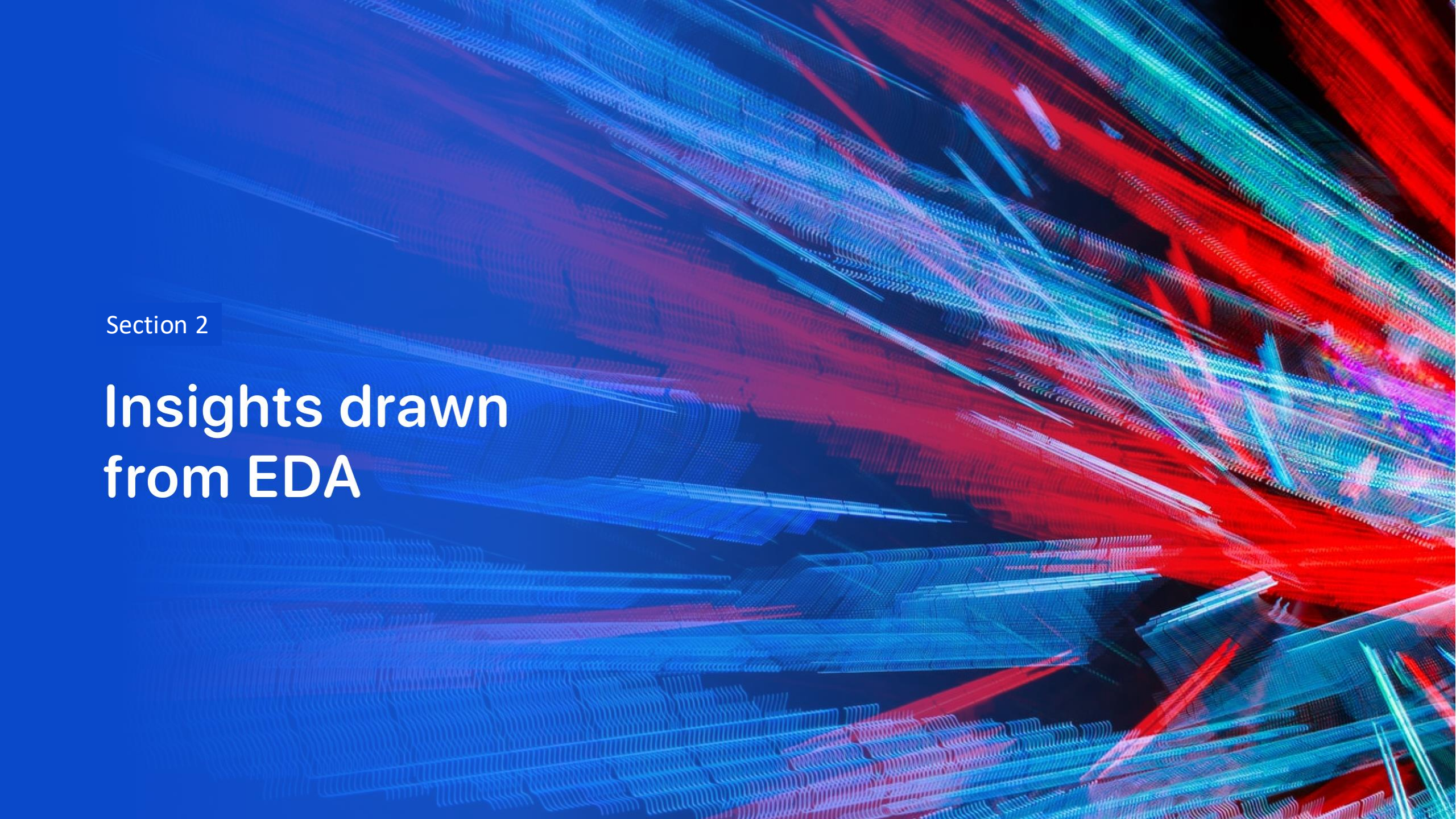
- Add dropdown list to enable launch site selection
- Add pie chart to show the total successful launches count for all sites and the success vs. failed counts
- Add a scatter chart of payload mass vs. success rate of different booster versions

Predictive Analysis (Classification)



Results

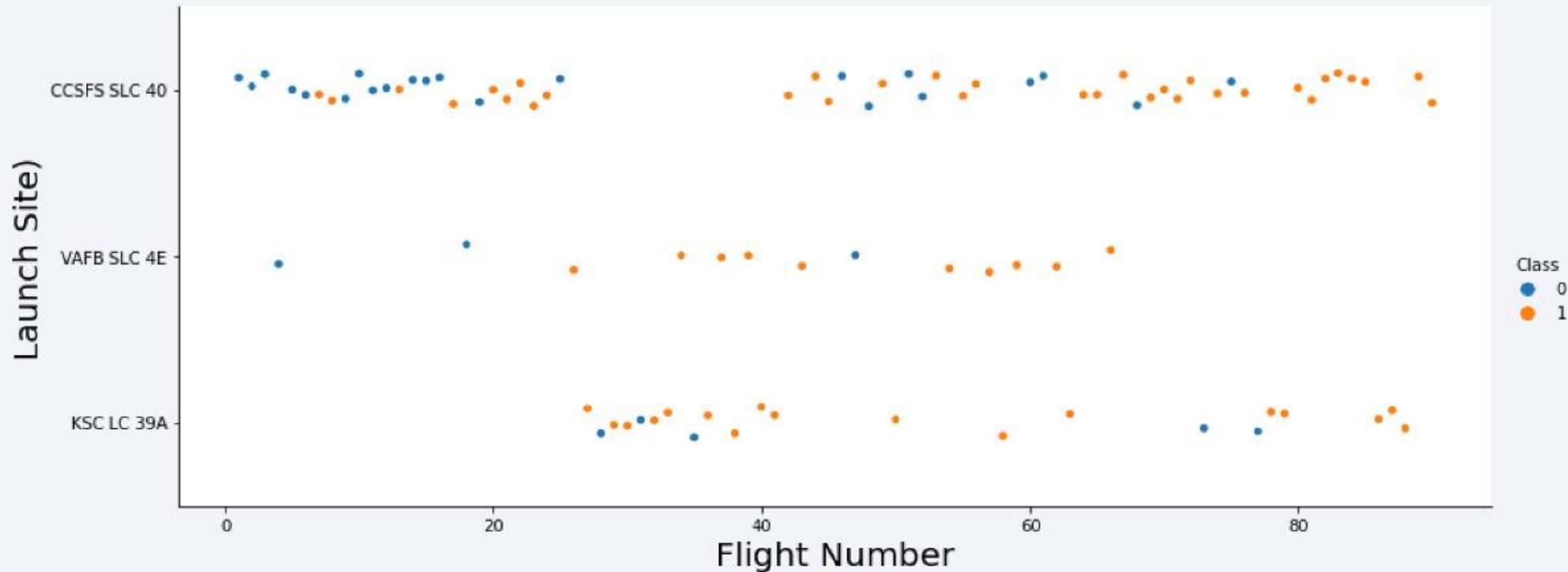
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

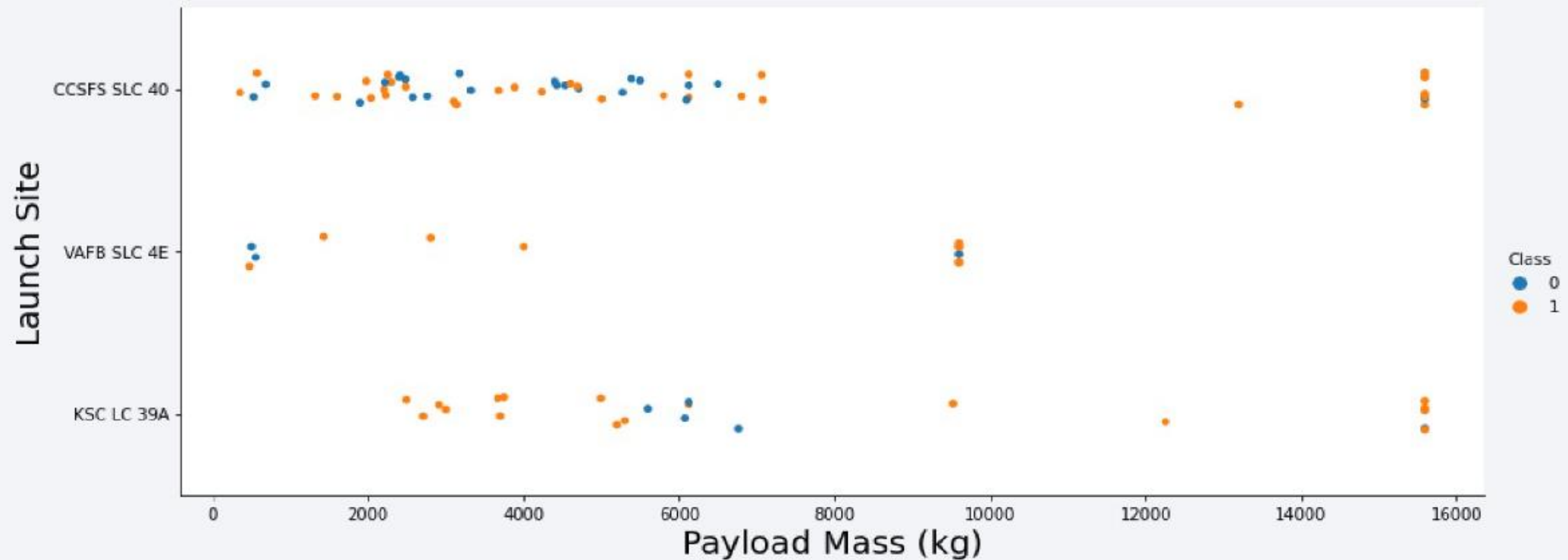
Insights drawn from EDA

Flight Number vs. Launch Site



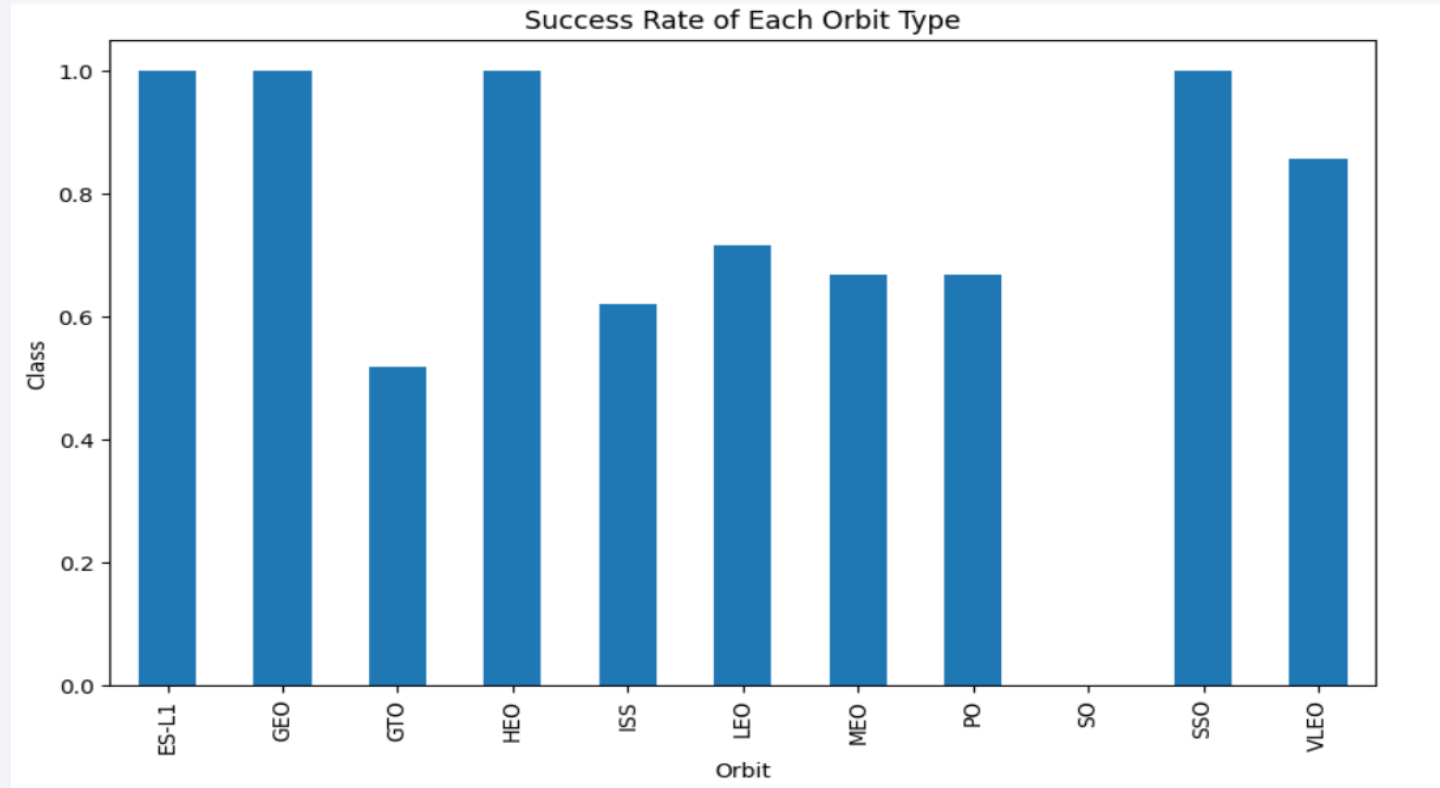
- Most flights were launched from CCAFS SLC 40
- The VAFB SLC 4E and KSC LC 39A sites have higher success rate than CCAFS SLC 40
- Newer flights have higher success rates than older flights

Payload vs. Launch Site



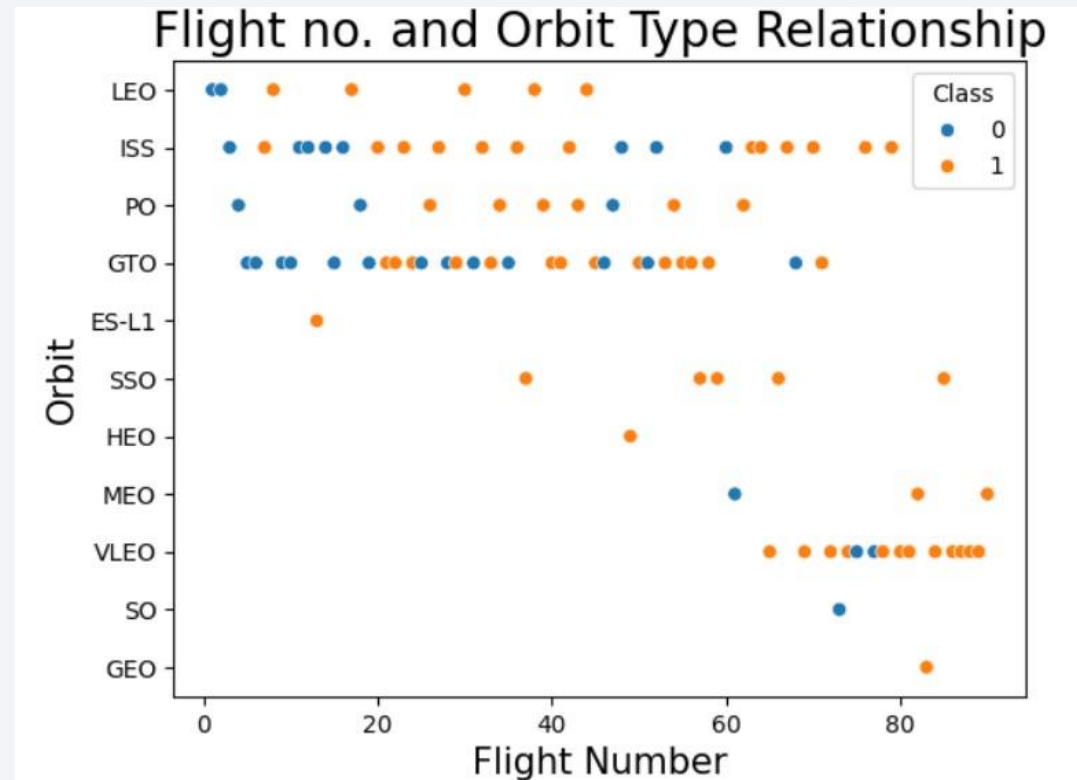
- Almost all flights with payload mass above 7000 Kg were successful.
- KSC LC 39A success rate for payload mass under 5500 kg is 100%.

Success Rate vs. Orbit Type



- The SO orbit has 0% success rate.
- The ES-L1, GEO, HEO and SSO orbits have 100% success rate.

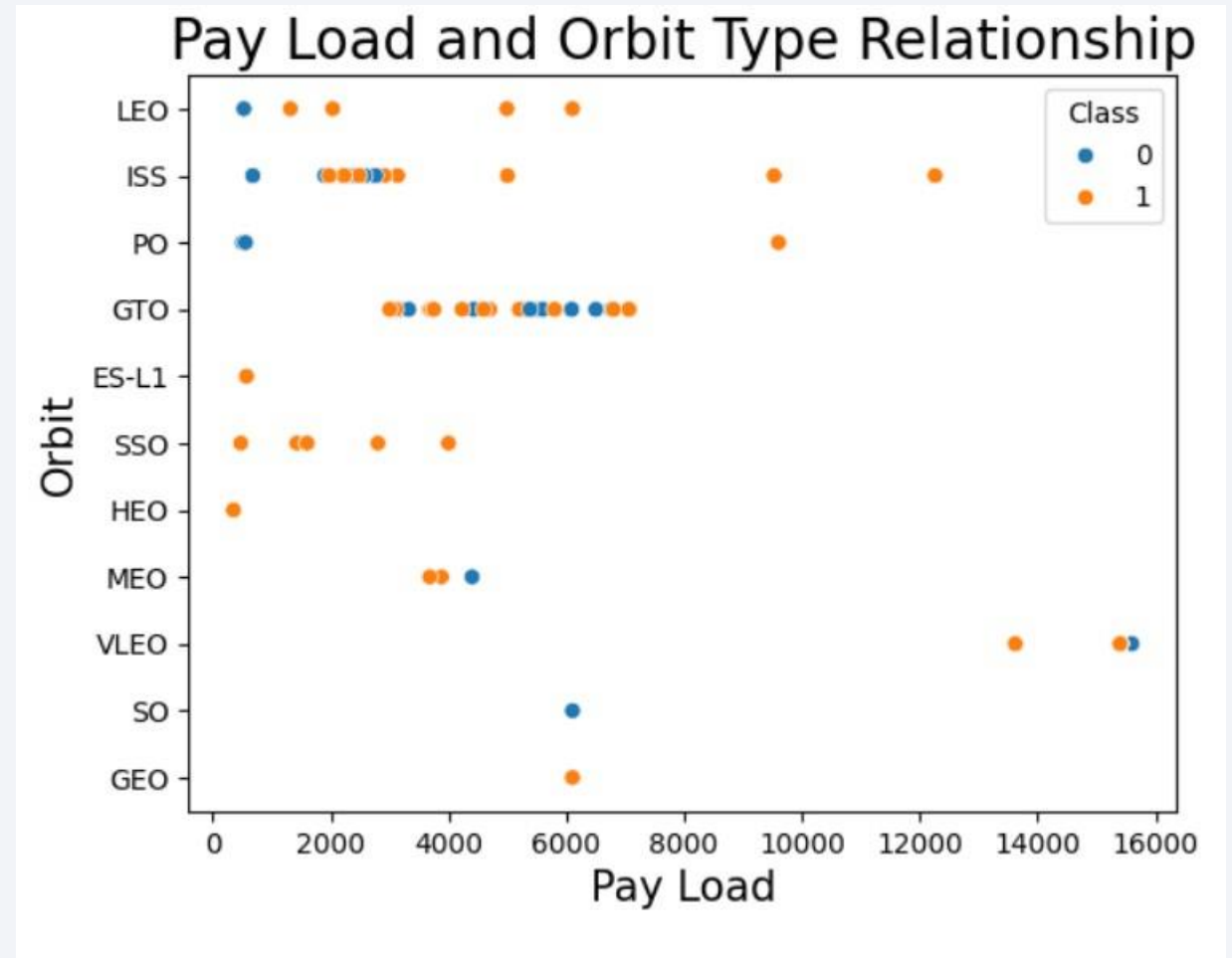
Flight Number vs. Orbit Type



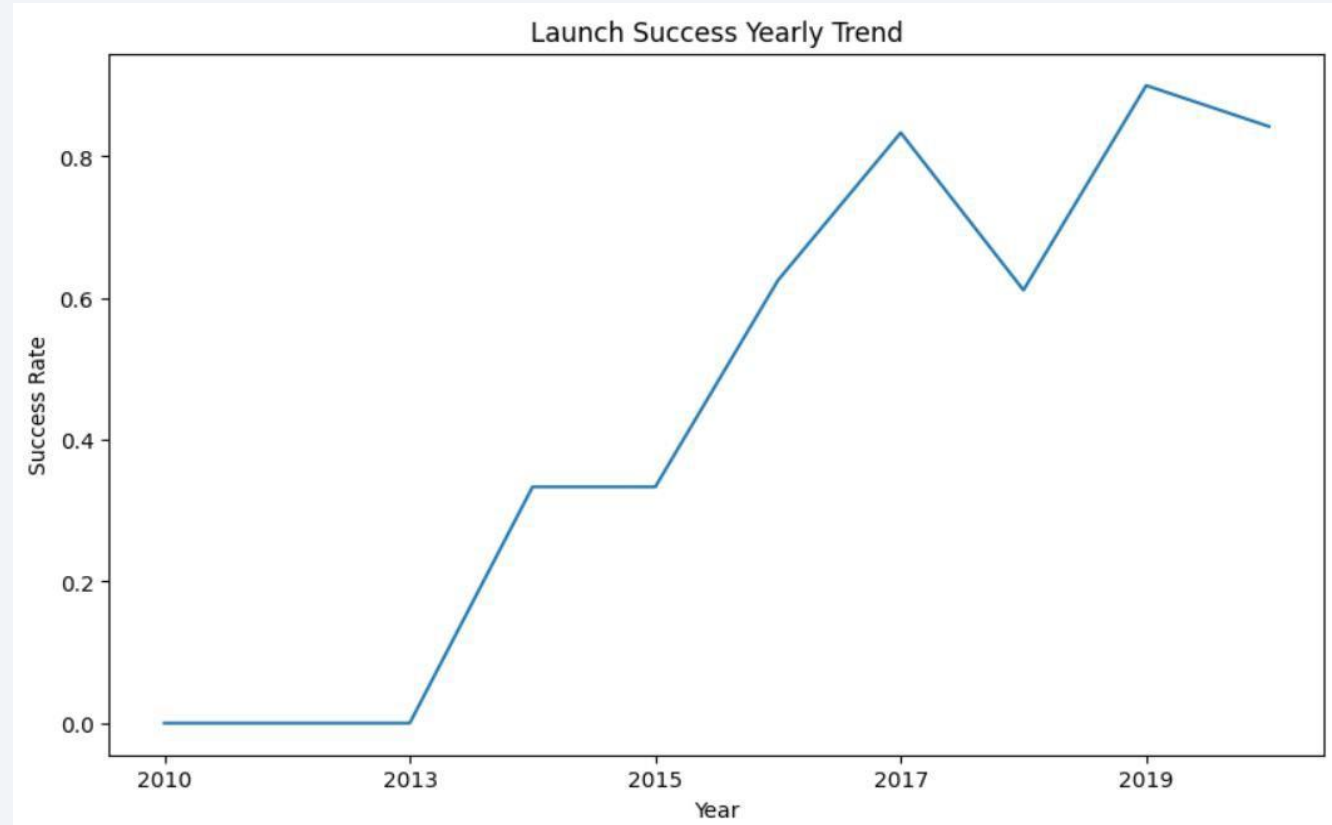
- The majority of the flights were launches to the ISS and GTO orbits.
- The data suggest no relationship between flight number and orbit type.

Payload vs. Orbit Type

- Payload masses above 10,000 Kg were placed in PO, ISS, and LEO orbits.
- Payloads above 4,000 and less than 8,000 Kg were placed in the GTO orbit.



Launch Success Yearly Trend



- The launches success rate increased steadily since 2013.
- During 2018 there was a drop in the launch success rates.

All Launch Site Names



```
1 features= df[['LaunchSite']]  
2 features.head()
```



LaunchSite



0	CCAFS SLC 40
1	CCAFS SLC 40
2	CCAFS SLC 40
3	VAFB SLC 4E
4	CCAFS SLC 40

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`

```
[ ] 1 %sql SELECT * FROM SPACEXTABLE WHERE launch_site LIKE 'CCA%' LIMIT 5;
```

```
➔ * sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The calculated total payload mass carried by boosters from NASA site = 48,213 Kg

```
%sql select SUM(PAYLOAD_MASS_KG_) from SPACEXTABLE where "Customer" like 'NASA (CRS)%'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
SUM(PAYLOAD_MASS_KG_)
```

```
48213
```

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is 2,535 Kg

```
%sql select AVG(PAYLOAD_MASS_KG_) from SPACEXTABLE where "Booster_Version" LIKE 'F9 v1.1%'
* sqlite:///my_data1.db
Done.
AVG(PAYLOAD_MASS_KG_)
2534.6666666666665
```

First Successful Ground Landing Date

- The date of the first successful landing outcome on ground pad was December 15th, 2015

```
[ ] 1 %sql SELECT MIN(Date) AS first_successful_landing_date FROM SPACEXTABLE WHERE landing_outcome = 'Success (ground pad)';
```

```
↳ * sqlite:///my_data1.db  
Done.  
first_successful_landing_date  
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[ ] 1 %sql SELECT Booster_Version FROM SPACEXTABLE WHERE landing_outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
[ ] 1 %sql SELECT mission_outcome, COUNT(*) AS total_count FROM SPACEXTABLE WHERE mission_outcome IN ('Success', 'Failure (in flight)') GROUP BY mis  
2
```



```
* sqlite:///my_data1.db
```

```
Done.
```

```
Mission_Outcome total_count
```

```
Failure (in flight)    1
```

```
Success                98
```

Boosters Carried Maximum Payload

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
[ ] 1 %sql SELECT booster_version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX (PAYLOAD_MASS__KG_) FROM SPACEXTABLE);  
2
```

```
➔ * sqlite:///my_data1.db
```

Done.

Booster_Version

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
[ ] 1 %sql SELECT CASE WHEN substr(Date, 6, 2) = '01' THEN 'January' WHEN substr(Date, 6, 2) = '02' THEN 'February' WHEN substr(Date, 6, 2) = '03' TH
2
```



* sqlite:///my_data1.db

Done.

month_name	Booster_Version	Launch_Site	Landing_Outcome
January	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
April	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
[ ] 1 %sql SELECT landing_outcome, COUNT(*) AS outcome_count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY landing_outcc
```

```
2
```

```
➔ * sqlite:///my_data1.db
```

Done.

Landing_Outcome	outcome_count
-----------------	---------------

No attempt	10
------------	----

Success (drone ship)	5
----------------------	---

Failure (drone ship)	5
----------------------	---

Success (ground pad)	3
----------------------	---

Controlled (ocean)	3
--------------------	---

Uncontrolled (ocean)	2
----------------------	---

Failure (parachute)	2
---------------------	---

Precluded (drone ship)	1
------------------------	---

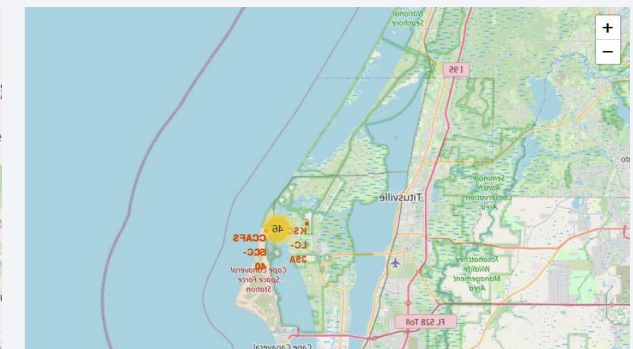
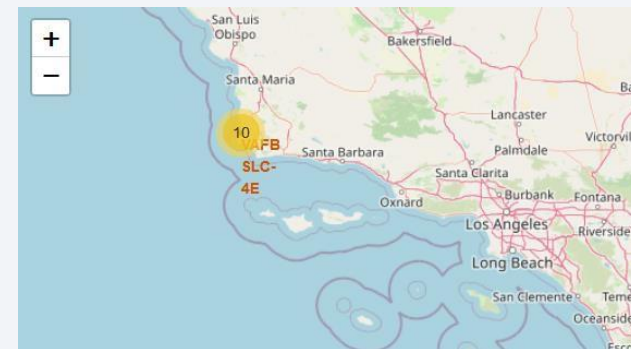
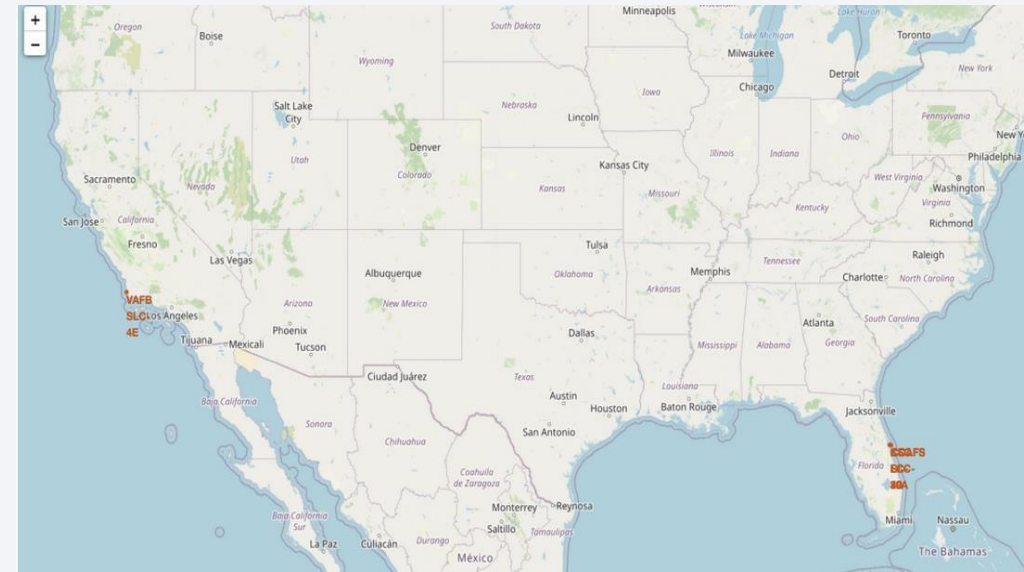
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue space with stars. The Earth's surface is dark blue, with bright yellow and orange lights from cities and towns. The lights are concentrated in the lower right quadrant of the image, following the curve of the Earth. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

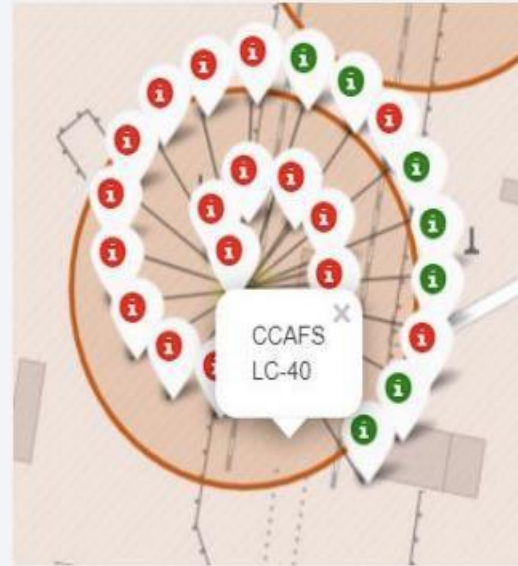
Launch Sites in California and Florida

- All launch sites considered in this project are in close proximity to the coast launching the ocean we minimize the risk of having any debris dropping on land.

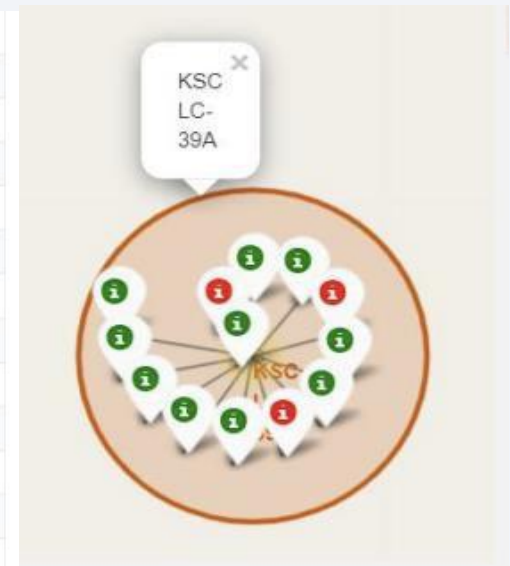


Color Labels Showing the Launch Sites on a Map

	Launch Site	Lat	Long	marker_color
0	CCAFS LC-40	28.562302	-80.577356	red
1	CCAFS LC-40	28.562302	-80.577356	red
2	CCAFS LC-40	28.562302	-80.577356	red
3	CCAFS LC-40	28.562302	-80.577356	red
4	CCAFS LC-40	28.562302	-80.577356	red
5	CCAFS LC-40	28.562302	-80.577356	red
6	CCAFS LC-40	28.562302	-80.577356	red
7	CCAFS LC-40	28.562302	-80.577356	red
8	CCAFS LC-40	28.562302	-80.577356	red
9	CCAFS LC-40	28.562302	-80.577356	red
10	CCAFS LC-40	28.562302	-80.577356	red
11	CCAFS LC-40	28.562302	-80.577356	red
12	CCAFS LC-40	28.562302	-80.577356	red
13	CCAFS LC-40	28.562302	-80.577356	red
14	CCAFS LC-40	28.562302	-80.577356	red
15	CCAFS LC-40	28.562302	-80.577356	red
16	CCAFS LC-40	28.562302	-80.577356	red
17	CCAFS LC-40	28.562302	-80.577356	green
18	CCAFS LC-40	28.562302	-80.577356	green
19	CCAFS LC-40	28.562302	-80.577356	red
20	CCAFS LC-40	28.562302	-80.577356	green



36	KSC LC-39A	28.573255	-80.646895	green
37	KSC LC-39A	28.573255	-80.646895	red
38	KSC LC-39A	28.573255	-80.646895	green
39	KSC LC-39A	28.573255	-80.646895	green
40	KSC LC-39A	28.573255	-80.646895	red
41	KSC LC-39A	28.573255	-80.646895	green
42	KSC LC-39A	28.573255	-80.646895	green
43	KSC LC-39A	28.573255	-80.646895	red
44	KSC LC-39A	28.573255	-80.646895	green
45	KSC LC-39A	28.573255	-80.646895	green
46	KSC LC-39A	28.573255	-80.646895	green
47	KSC LC-39A	28.573255	-80.646895	green
48	KSC LC-39A	28.573255	-80.646895	green

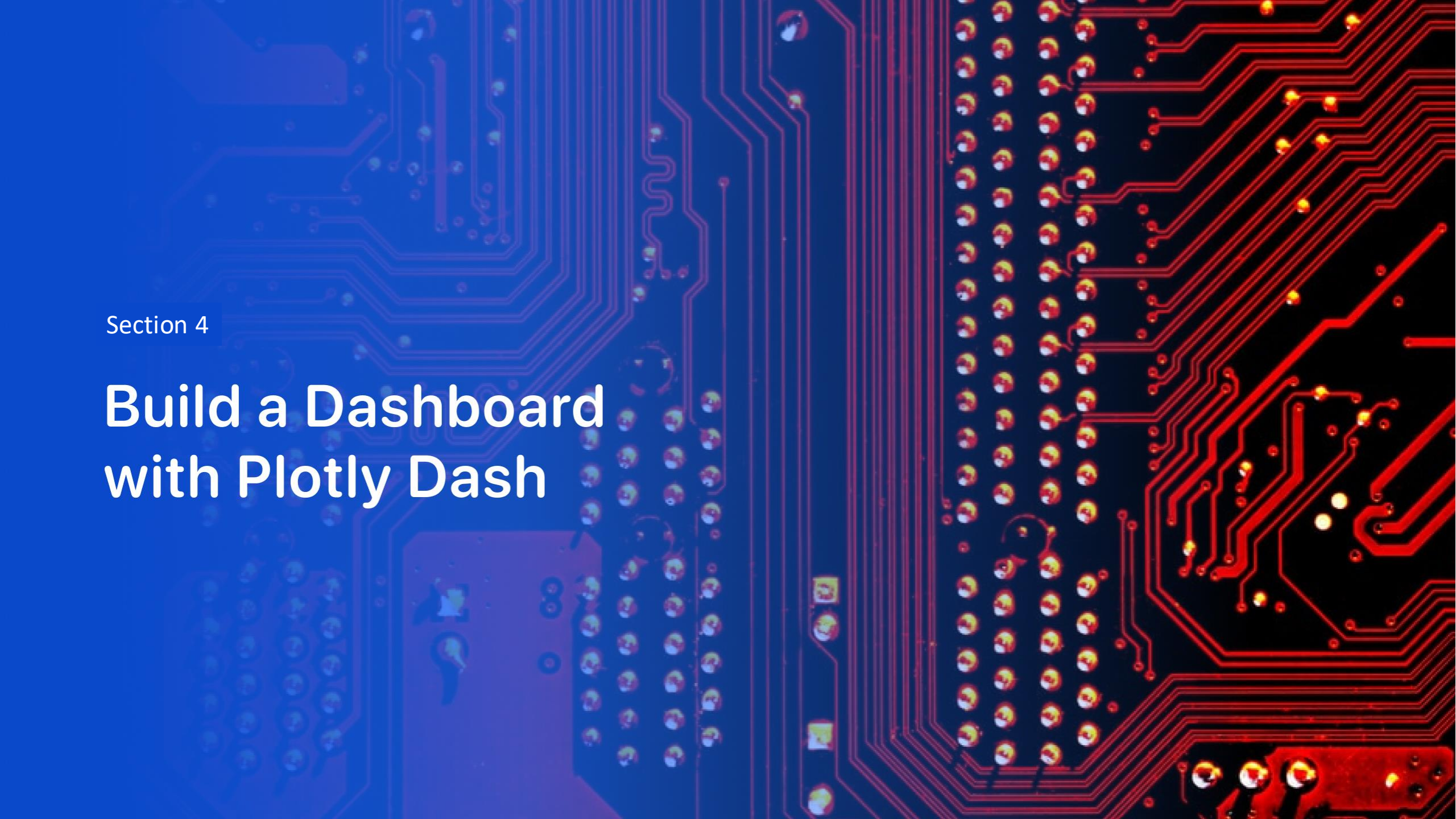


Success

Failure

Safe distance to launch site

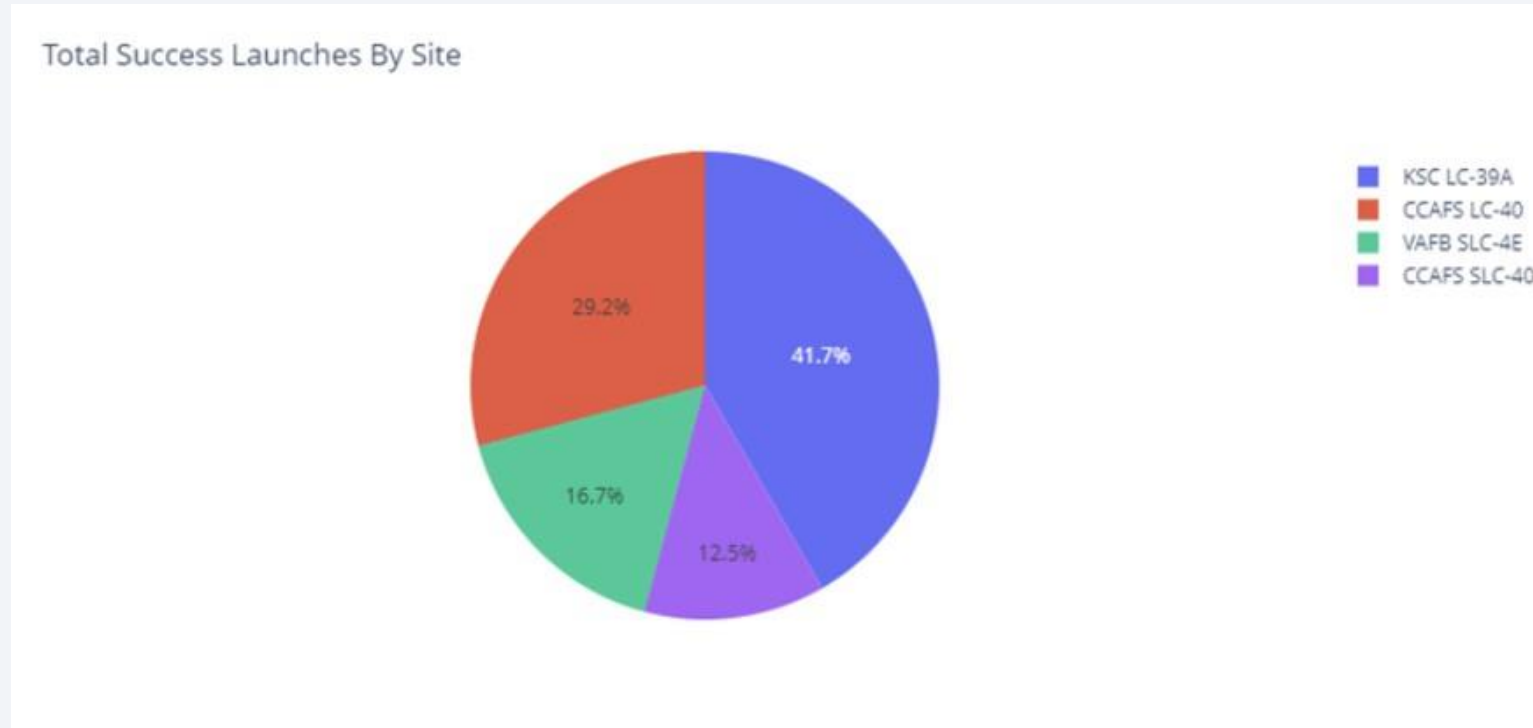




Section 4

Build a Dashboard with Plotly Dash

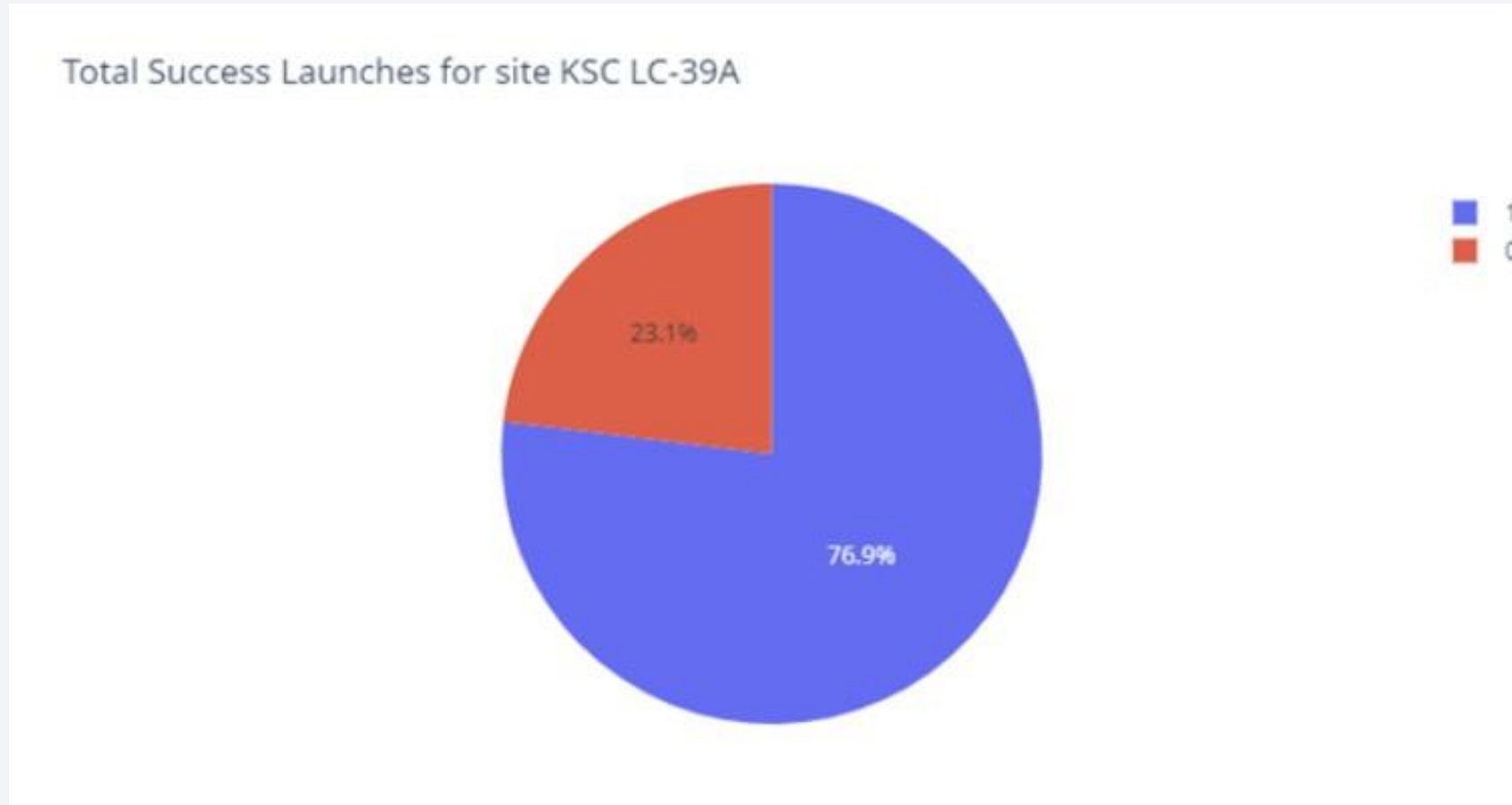
Launch success rate per site



The highest success launch rates were recorded at these sites :

1. KSC LC-39A (41.7%)
2. CCAFS LC-40 (29.2%)

KSC LC-39A launch success rate



Site KSC LC-39 success rate is 76.9%

Payload vs. Launch Outcome

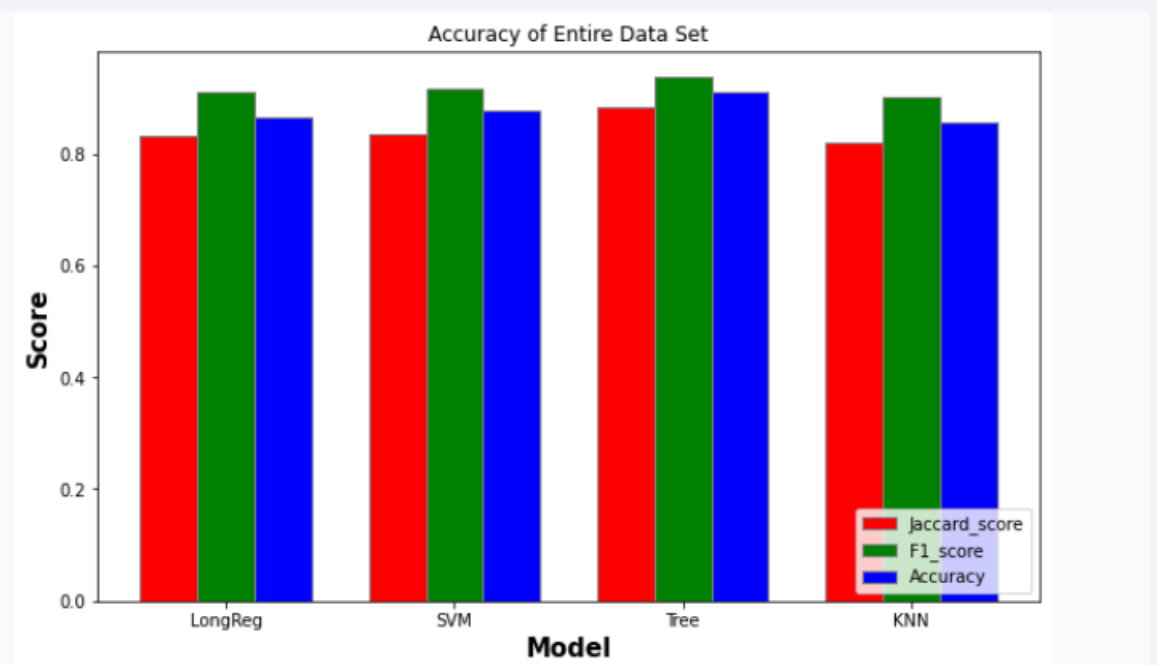
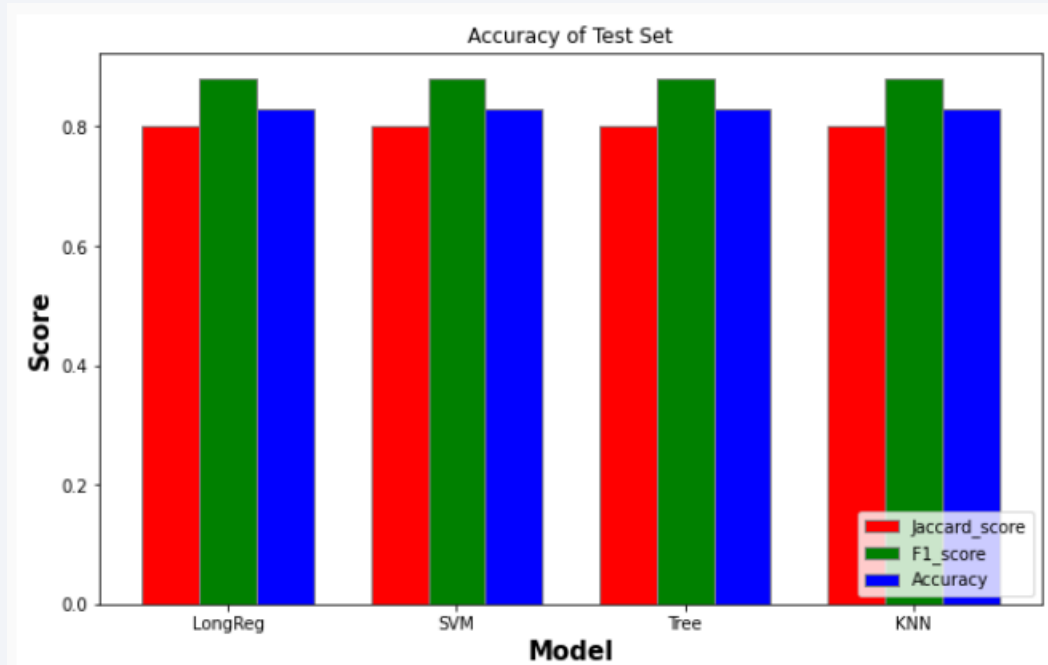
- Payloads is between 2000 and 5500 Kgs had the highest success rate



Section 5

Predictive Analysis (Classification)

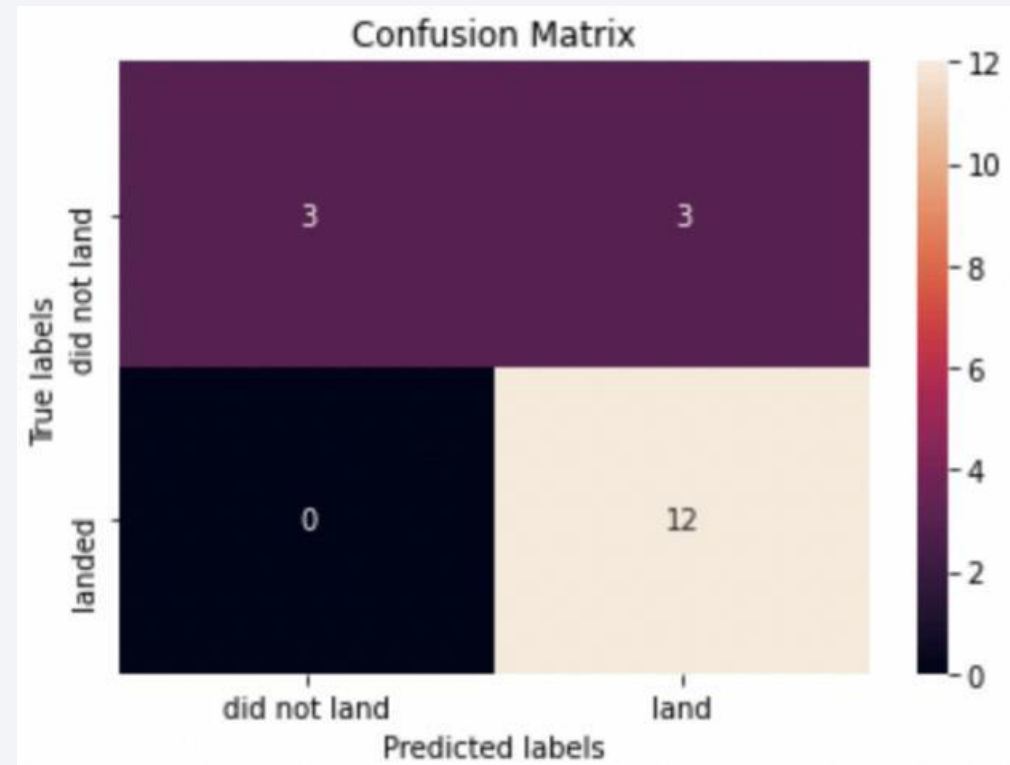
Classification Accuracy



- The Tree Model provided the best accuracy results for the entire data set.

Confusion Matrix

- The confusion matrix predicts:
 - 13 true positives
 - 3 false positives
 - 3 true positive
 - 0 false negative.



Conclusions

- The success rate for the rocket launches increased after 2013.
- Orbits GEO, HEO, ES-L1 and SSO have 100% launch success rate.
- Launch site KSC LC-39A has the highest success rate.
- The Decision Tree model is the best ML algorithm for analyzing the SpaceX data set and provided the best accuracy results.

Appendix

<https://github.com/Jhlberger/Jhlberger>

https://www.coursera.org/professional-certificates/ibm-data-science?campaignid=1876641588&adgroupid=70740725700&device=c&keyword=ibm%20data%20science%20professional%20certificate&matchtype=b&network=g&devicemodel=&adposition=&creativeid=347445112274&hide_mobile_promo=&gads_source=1

Thank you!

